FINM3013 Introduction to Financial Derivatives

Chapter 3 Exercises

Solutions

1.

The excess of the spot over the futures at the time the hedge is closed out is \$0.20 per ounce. If the trader is hedging the purchase of silver, the price paid is the futures price plus the basis. The trader therefore loses 60×5,000×\$0.20=\$60,000. If the trader is hedging the sales of silver, the price received is the futures price plus the basis. The trader therefore gains \$60,000.

2.

a) The company should short

$$\frac{(1.2-0.5)\times 100,000,000}{1000\times 250}$$

or 280 contracts.

b) The company should take a long position in

$$\frac{(1.5-1.2)\times 100,000,000}{1000\times 250}$$

or 120 contracts.

3.

a)

The number of contracts the fund manager should short is

$$0.87\times\frac{50,000,000}{1259\times 250}=138.20$$

Rounding to the nearest whole number, 138 contracts should be shorted.

b) The following table shows that the impact of the strategy. To illustrate the calculations in the table consider the first column. If the index in two months is 1,000, the futures price is 1000×1.0025. The gain on the short futures position is therefore

$$(1259-1002.50)\times 250\times 138=\$8,849,250$$

The return on the index is $3\times 2/12$ =0.5% in the form of dividend and

$-250/1250 = -20\%$ in the form of capital gains. The total return on the index is therefore -19.5%. The risk-free rate is 1% per two months. The return is therefore -20.5% in excess of the risk-free rate. From the capital asset pricing model we expect the return on the portfolio to be $0.87 \times -20.5\% = -17.835\%$ in excess of the risk-free rate. The portfolio return is therefore -16.835%. The loss on the portfolio is $0.16835 \times 50{,}000{,}000$ or \$8,417,500. When this is combined with the gain on the futures the total gain is \$431,750.

Index now	1250	1250	1250	1250	1250
Index Level in Two Months	1000	1100	1200	1300	1400
Return on Index in Two Months	-0.20	-0.12	-0.04	0.04	0.12
Return on Index incl divs	-0.195	-0.115	-0.035	0.045	0.125
Excess Return on Index	-0.205	-0.125	-0.045	0.035	0.115
Excess Return on Portfolio	-0.178	-0.109	-0.039	0.030	0.100
Return on Portfolio	-0.168	-0.099	-0.029	0.040	0.110
Portfolio Gain	-8,417,500	-4,937,500	-1,457,500	2,022,500	5,502,500
Futures Now	1259	1259	1259	1259	1259
Futures in Two Months	1002.50	1102.75	1203.00	1303.25	1403.50
Gain on Futures	8,849,250	5,390,625	1,932,000	-1,526,625	-4,985,250
Net Gain on Portfolio	431,750	453,125	474,500	495,875	517,250